HDOC Day-69

Recursive Fibonacci Series and Recursive Call Count

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Problem Statement

Write a recursive function to generate the first **n Fibonacci numbers**. Also count the number of recursive calls made.

C Program

```
#include <stdio.h>

long long calls = 0;

int fibonacci(int n)
{
calls++;
if (n == 0)
return 0;
if (n == 1)
return 1;
return fibonacci(n - 1) + fibonacci(n - 2);
}

int main()
{
int n, i;
printf("Enter number of terms: ");
scanf("%d", &n);

printf("Fibonacci series: ");
for (i = 0; i < n; i++)
printf("%d ", fibonacci(i));

printf("\nTotal recursive calls = %lld\n", calls);
return 0;
}
```

Simple Logic

The function **fibonacci(n)** calls itself to calculate the previous two Fibonacci values. Every time the function starts, **calls++** increases the recursive-call counter. The base cases are F(0) = 0 and F(1) = 1.

Kali Linux Terminal - Compilation and Actual Run

```
(kali@kali)-[~]
$ gcc hdoc69.c -o hdoc69

(kali@kali)-[~]
$ ./hdoc69
Enter number of terms: 5
Fibonacci series: 0 1 1 2 3
Total recursive calls = 19

(kali@kali)-[~]
$ ./hdoc69
Enter number of terms: 7
Fibonacci series: 0 1 1 2 3 5 8
Total recursive calls = 53

(kali@kali)-[~]
```

Recreated Kali-terminal-style snippet showing compilation and program execution.

Test Cases

n	Expected Fibonacci Series	Recursive Calls
1	0	1
5	0 1 1 2 3	19
7	0 1 1 2 3 5 8	53

Result

The first n Fibonacci numbers were generated successfully using recursion, and the total number of recursive function calls was counted.